Math for Dotter

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1 The Continuous One Bit Channel

1.1 Two Interpretations

The continuous one bit channel can be conceived in two ways: as a continuous channel with a one bit range, or as a series of times of identical events. The equivalency follows from the fact that the continuous channel may be characterized by the times at which its signal changes. To have a finite (differential) capacity we must constrain the channel to a minimum duration of signal, or equivalently, a minimum time between events. For our purpose, we prefer the latter interpretation of the one bit channel, since the user communicates using a sequence of identical events such as blinks or taps.

1.2 The Ideal Timing Distribution

What then, is the capacity of the one bit channel? We prefer the event interpretation for analysis. We consider the ideal distribution of the duration between events. Let T be the minimum duration between events and X be the additional delay beyond the minimum, so that the total duration between events is $T+X$. Let $A = T + E[X]$ be the expected inter-arrival time. The rate is given by

$$C = \max_{\mathcal{P}_X} \frac{h(X)}{A}.$$

We refer to the rate-maximizing distribution of X, $\mathcal{P}_X$, as the ideal timing distribution. We will proceed to show that the ideal timing distribution is an exponential distribution.

We treat the capacity, expected value, and differential entropy as functionals of the probability density function $f(x)$. To find the optimal distribution, we introduce a Lagrange multiplier μ for the normalization constraint $\int f(x)dx = 1$ and define the functional $J(f)$:

$$J(f) = \frac{h(X)}{A} - \mu \left(\int f(x)dx - 1 \right)$$

We maximize $J(f)$ by setting the variational derivative with respect to $f(x)$ to zero:

$$\frac{\delta J}{\delta f(x)} = 0$$

Using the quotient rule for variational derivatives, and noting that $\frac{\delta h(X)}{\delta f(x)} = -1 - \ln f(x)$ and $\frac{\delta A}{\delta f(x)} = \frac{\delta E[X]}{\delta f(x)} = x$, we have:

$$\frac{\delta}{\delta f(x)} \left(\frac{h(X)}{A} \right) = \frac{A(-1 - \ln f(x)) - h(X)x}{A^2}$$

Substituting this into the stationarity condition:

$$\frac{A(-1 - \ln f(x)) - h(X)x}{A^2} - \mu = 0$$

Solving for $f(x)$:

$$f(x) \propto \exp\left(-\frac{h(X)}{A}x\right)$$

proving that X is exponentially distributed with rate parameter equal to the rate of the channel (i.e., $\lambda^* = C$). This fact lets us solve for the rate,

$$C = \frac{1 - \ln C}{T + 1/C}$$

which by algebra has the solution,

$$C = W(T)/T$$

, where W is the Lambert W function.

1.2.1 A Geometric Intuition

The above result (that the ideal timing distribution is exponential) may be appreciated more readily with a geometric argument. We consider a large sample, $\mathbf{X}$, in N-dimensional space, representing the sequence of additional delays X_i. By the law of large numbers, its L_1 norm, $\|\mathbf{X}\|_1 = \sum_i X_i$, must be tightly distributed about the N-simplex, $\|\mathbf{X}\|_1 = NE[X]$. This can be thought of as the permissible region for $\mathbf{X}$. Thus the entropy maximizing distribution for $\mathbf{X}$ should assign a uniform distribution over the N-simplex, which is precisely what the exponential distribution does as it depends only on the L_1 norm, $f(\mathbf{X}) \propto \exp(-\lambda\|\mathbf{X}\|_1)$.

(Incidentally, this conception also provides a geometric proof of the fact that the differential entropy of the exponential distribution is $1 - \ln\lambda$, since the entropy of a uniform distribution over the N-simplex should be the logarithm of its volume. Since its side length is N/λ, its volume is $(N/\lambda)^N/N!$, and its entropy per sample (by Stirling's approximation) is

$$h(X) = \lim_{N\to\infty} \frac{1}{N}\left(N(\log(N - \log\lambda) - N(\log N - 1))\right) = 1 - \ln\lambda$$

as expected.)

2 Periodic Signals

2.1 Random Scan Times

2.2 Periodic signals

2.3 Approximating A Uniform Distribution with a Gaussian Mixture

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We have considered the ideal timing distribution and the periodic timing distribution. How can we induce a uniform distribution on the user's signal? There are several targets that the user may aim for, corresponding to all the prefixes displayed. Each target has an assigned signal time, μ_i and a fixed and known (mixing) probability, π_i. Given the noise $\mathcal{E} \sim \mathcal{N}(0, \sigma)$, the distribution of the observed signal is a mixture of gaussians with means mu_i and constant variance σ^2. Given the mixing probabilities, π_i, how should we position the μ_i to maximize the entropy of the signal?

Intuitively we want to space apart the μ_i to approximate a uniform distribution. Those components with higher pi_i should receive more space.

As soon as we receive a signal we recompute the posterior probabilities of the targets and must assign new phases to them. Hence this computation is on the critical path and ought to be computed in a few milliseconds. This rules out heavier techniques like gradient descent on a numerical approximation of the entropy.

We discuss two heuristic approaches to the problem as well as an approximation for the entropy which may computed efficiently.

2.3.1 Two Heuristics

UNFINISHED We wish to space the components of the mixture as to approximate a uniform distribution. One heuristic is to determine (by binary search) a value C such that

$$\sum_i 2w_i = P$$

where the widths allotted to each component, w_i, are given by

$$w_i = \mathrm{Re}\left(\sigma\sqrt{2\left(\ln\left(\frac{\pi_i}{\sqrt{2\pi}\sigma}\right) - C\right)}\right)$$

where P is the length of the interval. The means are then given by $\mu_i = w_i + \sum_{j=1}^{i-1} 2w_j$.

This heuristic is derived from approximating the entropy of the mixture as $h(X) \approx -\ln f_{\min}$ where $f_{\min}$ is the minimum value of the mixture density, and approximating the mixture by its greatest component at a given point,

$$f(x) \approx \max_i f_i(x)$$

The value of C represents the minimum log density of the approximated mixture, i.e., the minimum over the interval of the maximum log density of the components. The width $2w_i$ is the range over which the ith component may provide a log density greater than C. If $\ln f_i(\mu_i) < C$, where $f_i(\mu_i)$ denotes the maximum density of the ith component at $x = 0$, then clearly $w_i = 0$, otherwise it is as given above. Our algorithm thus maximizes the minimum density of the maximum component over the interval. Although somewhat crude, this heuristic is efficient.

A refinement may be made by instead approximating the mixture by the sum of its two greatest components, and continuing to try to maximize the minimum density of the distribution. In this case, adjacent components should be spaced apart such that the log of the sum of their densities attains a minimum value of C. Given two adjacent components, i and $i+1$, we are interested in their separation, l_i. Letting $\mu_i = 0$ and $\mu_{i+1} = l_i$, we may solve for the minimum value of the two-component mixture:

$$f(x) = \pi_i \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{x^2}{2\sigma^2}\right) + \pi_{i+1} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-l_i)^2}{2\sigma^2}\right)$$

Note that completing the square for the sum of squared distances gives:

$$(x-\mu_1)^2 + (x-\mu_2)^2 = 2\left(x - \frac{\mu_1+\mu_2}{2}\right)^2 + \frac{(\mu_1-\mu_2)^2}{2}$$

Taking the derivative with respect to x and setting it to zero we have,

$$\frac{d}{dx}f(x) = \frac{1}{\sqrt{2\pi\sigma^2}}\left(\frac{-2x\pi_i}{2\sigma^2}\exp\left(-\frac{x^2}{2\sigma^2}\right) - \frac{-2(x-l_i)\pi_{i+1}}{2\sigma^2}\exp\left(-\frac{(x-l_i)^2}{2\sigma^2}\right)\right) = 0$$

Algebra yields,

$$\ln\left(\frac{Rx}{l_i - x}\right) = \frac{xl_i}{\sigma^2} - \frac{l_i^2}{2\sigma^2}$$

, where $R = \pi_i/\pi_{i+1}$. Taken with the condition that $\ln f(x) = C$ the two variables, C and R, implicitly determine l_i. So, using a precomputed table of l_i values for a range of C and R values, we can proceed as before, using binary search to find the value of C such that $\sum_i l_i = P$. As before, we will exclude components with $\ln f_i(\mu_i) < C$ from calculations. As long as each adjacent component has a maximum log density greater than C, it will be possible to space them such that the log of their sum attains a minimum value of C.

2.3.2 Approximations to the entropy based on pairwise distances

In order to optimize the placement of the means to maximize entropy, we may minimize a proxy objective J based on pairwise distances between the components. We consider approximations to the entropy for both Gaussian and von Mises mixture models.

Gaussian Mixture Model with Rényi Entropy The Rényi 2-entropy, also known as collision entropy, is defined as $H_2(X) = -\ln \int p(x)^2 dx$. Maximizing this entropy is equivalent to minimizing the information potential $J = \int p(x)^2 dx$. For a mixture of Gaussians $p(x) = \sum_{i=1}^{N} \pi_i \mathcal{N}(x|\mu_i, \sigma^2)$, the integral of the squared density is:

$$J = \int \left(\sum_{i=1}^{N} \pi_i \mathcal{N}(x|\mu_i, \sigma^2) \right)^2 dx = \sum_{i=1}^{N} \sum_{j=1}^{N} \pi_i \pi_j \int \mathcal{N}(x|\mu_i, \sigma^2) \mathcal{N}(x|\mu_j, \sigma^2) dx$$

The integral of the product of two Gaussians is itself a Gaussian evaluated at the difference of their means:

$$\int \mathcal{N}(x|\mu_i, \sigma^2) \mathcal{N}(x|\mu_j, \sigma^2) dx = \mathcal{N}(0|\mu_i - \mu_j, 2\sigma^2) = \frac{1}{\sqrt{4\pi\sigma^2}} \exp\left(-\frac{(\mu_i - \mu_j)^2}{4\sigma^2} \right)$$

By dropping the leading scalar and the self-interaction terms ($i = j$) since they are constant with respect to the optimal placement of the means, the objective to minimize simplifies to the pairwise sum:

$$J = \sum_{i<j} \pi_i \pi_j \exp\left(-\frac{(\mu_i - \mu_j)^2}{4\sigma^2} \right)$$

Gaussian Mixture Model with Kolchinsky Objective Kolchinsky and Tracey [1] introduced a family of computationally efficient estimators for mixture entropy that rely on calculating pairwise distances between component distributions, proving they form a strict lower bound on the true mixture entropy. For homoskedastic Gaussian mixtures, maximizing this lower bound utilizing the Bhattacharyya distance yields a similar pairwise objective. The effective variance in the penalty term is simply doubled compared to the Rényi objective:

$$J = \sum_{i<j} \pi_i \pi_j \exp\left(-\frac{(\mu_i - \mu_j)^2}{8\sigma^2} \right)$$

Von Mises Mixture Model with Rényi Entropy When the components are distributed on the unit circle as von Mises distributions with homoskedastic concentration κ, we again aim to minimize the integral of the squared density. The integral of the cross-term of two von Mises components evaluates to a modified Bessel function of the first kind of order zero, I_0. Stripping out constants and self-terms, the objective becomes:

$$J = \sum_{i<j} \pi_i \pi_j I_0 \left(2\kappa \cos\left(\frac{\mu_i - \mu_j}{2} \right) \right)$$

Von Mises Mixture Model with Kolchinsky Objective Applying Kolchinsky's lower bound using the Bhattacharyya coefficient to the von Mises mixture model yields an objective with an identical structure to the Rényi case. The difference lies solely in the concentration parameter, which is halved:

$$J = \sum_{i<j} \pi_i \pi_j I_0 \left(\kappa \cos\left(\frac{\mu_i - \mu_j}{2} \right) \right)$$

2.3.3 An approximation for the entropy via Cramér's theorem

UNFINISHED Here we discuss a closed form approximation for the entropy in terms of the mixing weights, π_i, and the chosen means, μ_i. This enables efficient gradient descent on the means μ_i. We take a geometric approach and consider a large sample, $\mathbf{X}$, in N-dimensional space. Let A_i be a discrete random variable taking on the values of μ_i, with respective probabilities π_i. Let $\mathcal{E}_i \sim \mathcal{N}(0, \sigma^2)$ be random noise on the ith component. Then we have that

$$\mathbf{X} = \mathbf{A} + \mathcal{E}$$

And

$$\begin{aligned}
H(\mathbf{X}) &= H(\mathbf{A},\mathbf{X}) - H(\mathbf{A}|\mathbf{X}) \\
&= H(\mathbf{A},\mathcal{E}) - H(\mathbf{A}|\mathbf{X}) \quad \text{If } \mathbf{A} \text{ is fixed, } \mathbf{X} \text{ and } \mathcal{E} \text{ differ only be a translation} \\
&= H(\mathcal{E}) + H(\mathbf{A}) - H(\mathbf{A}|\mathbf{X}) \\
&= H(\mathcal{E}) + I(\mathbf{A};\mathbf{X})
\end{aligned}$$

Since the entropy of the noise is constant, our objective is to maximize the mutual information between $\mathbf{X}$ and the component choice, $\mathbf{A}$. We frame this in terms of trying to infer the true component choice, $\mathbf{A}$, from the observed sample, $\mathbf{X}$. To this end, we consider the prior distribution of the vertices' distances to the sample, $D = \|\mathbf{X} - \mathbf{B}\|_2^2$, where $\mathbf{B}$ represents a candidate component choice and is distributed identically to $\mathbf{A}$. Noting that, $D = \sum_i D_i$ where $D_i = (A_i + \mathcal{E}_i - B_i)^2$, we have that

$$P(\bar{D} = d) \approx \exp\left(-NI(d)\right)$$

and the likelihood for $\bar{D}$ is

$$P(\mathbf{X}|\bar{D} = d) = \exp\left(-\frac{1}{2\sigma^2} Nd\right)$$

Thus the posterior mode of $\bar{D}$ is attained where $N(I(d) + \frac{d}{2\sigma^2})$ is minimized, which occurs when $I'(d) = -\frac{1}{2\sigma^2}$. At this point, the prior probability of the posterior mode will be $\exp(-NI(d))$, meaning that the observation delivers $NI(d)$ bits of information about the component choice, or $I(d)$ bits per sample.

$I(d)$ had a closed form expression, although its derivation is a somewhat formidable piece of algebra. We begin by defining the random variable $C_i = A_i - B_i$, which is also a discrete random variable taking on the values of $d_{ij} = \mu_i - \mu_j$ with probabilities $\rho_{ij} = \pi_i \pi_j$.

Solving for I in terms of its derivative, and setting it to $-\frac{1}{2\sigma^2}$, we have that

$$\mathbf{I}\left(\theta = -\frac{\mathbf{1}}{\mathbf{2}\sigma^{\mathbf{2}}}\right) = \frac{\mathbf{1}}{\mathbf{2}}\ln(\mathbf{2}) - \frac{\mathbf{1}}{\mathbf{4}} - \ln(\mathbf{Z}) - \frac{\mathbf{1}}{\mathbf{8}\sigma^{\mathbf{2}}}\frac{\mathbf{Y}}{\mathbf{Z}}$$

where

$$\mathbf{Z} = \sum_{\mathbf{i},\mathbf{j}} \rho_{\mathbf{ij}} \exp\left(-\frac{\mathbf{d}_{\mathbf{ij}}^{\mathbf{2}}}{\mathbf{4}\sigma^{\mathbf{2}}}\right)$$

and

$$\mathbf{Y} = \sum_{\mathbf{i},\mathbf{j}} \rho_{\mathbf{ij}} \mathbf{d}_{\mathbf{ij}}^{\mathbf{2}} \exp\left(-\frac{\mathbf{d}_{\mathbf{ij}}^{\mathbf{2}}}{\mathbf{4}\sigma^{\mathbf{2}}}\right)$$

.

The final term may be recognized as the expected squared distance of C_i under a Boltzmann distribution.

2.3.4 An exact expression for the rate using Cramér's Theorem

UNFINISHED The above provides a lower bound on the rate. Here we discuss and correct its error. The error in the above stems from the fact that $X_i^{(t)}$ is constant across (t). We are interested in the distance distribution of the vertices B relative to a fixed sample. By treating $X_i^{(t)}$ as independently distributed across t we overestimate the variance of the distance distribution. Here we consider the special case of $K = 2$ and $\vec{\pi} = (1/2, 1/2)$. We are interested in the conditional entropy $H(B|X)$, since this tells us the mutual information up to a constant, $I(B;X) = H(B) - H(B|X)$. In this special case, we fix $\mu_1 = 0$. Then the vertices form a hypercube and without loss of generality we may choose $\mathbf{A}^{(0)} = \vec{0}$ and thus $\mathbf{X}^{(t)} = \mathcal{E}^{(t)}$. To facilitate the analysis we will discretize the noise distribution (in such a way as to preserve the likelihood in D) so that E_i realizes the value of e_z with probability $P(E_i = e_z)$. A "typical" sample of X then is equal to e_z in $NP(E_i = e_z)$ coordinates. So,

$$D = Nx = \sum_z \sum_i^{NP(E_i = e_z)} (B_i - e_z)^2$$

We choose N sufficiently large so that Cramér's law holds for each interior sum. Then let x_z denote the mean of the inner sum for z. Then,

$$g(\vec{x}) = x = \sum_z P(E_i = e_z) x_z \tag{1}$$

For given x, we will estimate its prior probability by the probability of the prior mode for $\vec{x}|x$. To determine this mode we treat x as a constant and solve for the x_z. So our constraint is as given in (1) and our objective it to maximize

$$f(\vec{x}) = \ln P(\vec{x}) = \sum_z -NP(E_i = e_z) I_z(x_z) \tag{2}$$

Setting $\nabla f = \lambda \nabla g$ we have that

$$\begin{aligned} -NP(E_i = e_z) I'(x_z) &= \lambda P(E_i = e_z) \\ I'(x_z) &= -\frac{\lambda}{N} \end{aligned}$$

and

$$\frac{d}{dx} f(x) = \lambda$$

And the posterior mode of the distance distribution will be attained where

$$\frac{d}{dx} f(x) = -\frac{d}{dx} \ln(\exp(-\frac{Nx}{2\sigma^2}))$$

Hence, $\lambda = \frac{N}{2\sigma^2}$ and

$$I_z'(x_z) = -\frac{1}{2\sigma^2}$$

Fortunately, we can explicitly state the value of $I_z(x_z)$ in terms of its derivative. Letting,

$$\begin{aligned} t^* &= I_z'(x_z) \\ Z &= \sum_i \pi_i e^{t^*(\mu_i - e_z)^2} \\ Y &= \sum_i \pi_i (\mu_i - e_z)^2 e^{t^*(\mu_i - e_z)^2} \\ I_z(t^* = \frac{-1}{2\sigma^2}) &= -\frac{1}{2\sigma^2} \frac{Y}{Z} - \ln(Z) \end{aligned}$$

After learning $D = x$ the remaining entropy in B, $H(B|D = x)$, is given by the log of its prior probability. From this we get that the mutual information is, up to a constant,

$$E_e \left[-\frac{1}{2\sigma^2} \frac{Y}{Z} - \ln(Z) \right]$$

where Y, Z depend on e. The latter half is the cross entropy between the noise and the mixture! (Actually this whole thing is obvious; it is the info remaining in B_i after learning the sample X_i).

2.3.5 An incorrect (but tantalizing) decomposition

Here we repeat the analysis, seeking an exact expression for the mutual information. Here, $A, \mathcal{E}, X$, etc. are 1-dimensional random variables. We define (meaningless) auxiliary random variables $B_1, B_2 \sim Discrete(\vec{\mu}, \vec{\pi})$ and $\mathcal{E}_1, \mathcal{E}_2 \sim \mathcal{N}(0, \sigma^2)$, with all four variables being independent in order to facilitate the writing of some expressions.

We start from the defintion of conditional entropy to obtain,

$$
\begin{aligned}
H(A|X) &= E_x\left[H(A|X=x)\right] \\
&= \int P(X=x)\left(\sum_a P(A=a|X=x)(-\ln P(A=a|X=x))\right)dx \\
&= \int P(X=x)\left(\sum_a P(A=a|X=x)\left(\ln P(X=x) - \ln P(A=a) - \ln P(X=x|A=a)\right)\right)dx \\
&= T_1 - T_2 - T_3
\end{aligned}
$$

where

$$
\begin{aligned}
T_1 &= \int P(X=x)\left(\sum_a P(A=a|X=x)\ln P(X=x)\right)dx \\
T_2 &= \int_x P(X=x)\left(\sum_a P(A=a|X=x)\ln P(A=a)\right)dx \\
T_3 &= \int_x P(X=x)\left(\sum_a P(A=a|X=x)\ln P(X=x|A=a)\right)dx
\end{aligned}
$$

which three terms we will treat separately. First,

$$
\begin{aligned}
T_1 &= \int P(X=x)\left(\sum_a P(A=a|X=x)\ln P(X=x)\right)dx \\
&= \int P(X=x)\ln P(X=x)\left(\sum_a P(A=a|X=x)\right)dx \\
&= \int P(X=x)\ln P(X=x)dx \\
&= -h(X)
\end{aligned}
$$

Second,

$$
\begin{aligned}
T_2 &= \int_x P(X=x)\left(\sum_a P(A=a|X=x)\ln P(A=a)\right)dx \\
&= \int_x P(X=x)\left(\sum_a \frac{P(A=a)P(X=x|A=a)}{P(X=x)}\ln P(A=a)\right)dx \\
&= \int_x \left(\sum_a P(A=a)P(X=x|A=a)\ln P(A=a)\right)dx \\
&= \int_x \left(\sum_{b_2} P(B_2=b_2)P(\mathcal{E}_2=x-b_2)\ln P(B_2=b_2)\right)dx \\
&= \sum_{b_1} P(B_1=b_1)\int_{e_1} P(\mathcal{E}_1=e_1)\sum_{b_2} P(B_2=b_2)P(\mathcal{E}_2=b_1+e_1-b_2)\ln P(B_2=b_2)de_1 \\
&= \sum_{b_1,b_2} P(B_1=b_1)P(B_2=b_2)\ln P(B_2=b_2)\int_{e_1} P(\mathcal{E}_1=e_1)P(\mathcal{E}_2=b_1+e_1-b_2)de_1 \\
&= \sum_{b_1,b_2} P(B_1=b_1)P(B_2=b_2)\ln P(B_2=b_2)\int_{e_1} P(\mathcal{E}_1=e_1)P(\mathcal{E}_2-\mathcal{E}_1=b_1-b_2|\mathcal{E}_1=e_1)de_1 \\
&= \sum_{b_1,b_2} P(B_1=b_1)P(B_2=b_2)\ln P(B_2=b_2)P(\mathcal{E}_2-\mathcal{E}_1=b_1-b_2) \\
&= \frac{1}{2}\sum_{b_1,b_2} P(B_1=b_1)P(B_2=b_2)\left[\ln P(B_1=b_1)+\ln P(B_2=b_2)\right]P(\mathcal{E}_2-\mathcal{E}_1=b_1-b_2)
\end{aligned}
$$

for which defining $\rho_{ij}=P(B_1=b_i)P(B_2=b_j)$ and $d_{ij}=b_i-b_j$, is reduced to,

$$T_2=\frac{1}{2}\sum_{i,j}\rho_{ij}\ln(\rho_{ij})P(\mathcal{E}_2-\mathcal{E}_1=d_{ij})$$

Third,

$$
\begin{aligned}
T_3 &= \int_x P(X=x)\left(\sum_a P(A=a|X=x)\ln P(X=x|A=a)\right)dx \\
&= \int_x P(X=x)\left(\sum_a \frac{P(A=a)P(X=x|A=a)}{P(X=x)}\ln P(X=x|A=a)\right)dx \\
&= \int_x \left(\sum_a P(A=a)P(X=x|A=a)\ln P(X=x|A=a)\right)dx \\
&= \sum_a P(A=a)\int_x P(X=x|A=a)\ln P(X=x|A=a)dx \\
&= \sum_a P(A=a)\int_x P(X=x|A=a)\ln P(X=x|A=a)dx \\
&= \sum_a P(A=a)\int_x P(\mathcal{E}_1=x-a)\ln P(\mathcal{E}_1=x-a)dx \\
&= \sum_a P(A=a)\int_{e_1=-\infty}^{\infty} P(\mathcal{E}_1=e_1)\ln P(\mathcal{E}_1=e_1)de_1 \\
&= \int_{e_1=-\infty}^{\infty} P(\mathcal{E}_1=e_1)\ln P(\mathcal{E}_1=e_1)de_1 \\
&= -h(\mathcal{E}_1)
\end{aligned}
$$

Hence,

$$
\begin{aligned}
H(A|X) &= T_1 - T_2 - T_3 \\
&= -T_2 - (-T_1 + T_3) \\
&= -\frac{1}{2}\sum_{i,j}\rho_{ij}\ln(\rho_{ij})P(\mathcal{E}_2-\mathcal{E}_1=d_{ij}) - (h(X)-h(\mathcal{E}_1))
\end{aligned}
$$

3 User Parameters

3.1 The Likelihood Model

We model our observations of the user's signal as being subject to two sources of noise: an outlier probability, ρ, that the signal was unintended, and additive noise on intended signals. We model the unintended signals as uniformly distributed, and the noise as a Gaussian. (We do not require that the noise have zero mean, since the user may be consistently late or early, relative to the intended time.) Hence our likelihood model for the received signal is a uniform-gaussian mixture model, characterized by the outlier probability, ρ, and the parameters of the noise, μ_ϵ, σ_ϵ.

3.2 Directional Statistics

3.3 The Cost of Unintended Signals

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The user's speed can be calculated from their performance parameters. Here we consider the effect of ρ, the probability of unintended signals. Given an observation, D, the gain in logits for the true hypothesis h is $\ln P(D|h) - \ln P(D|\neg h)$. When the posterior probability of h is small, this gain is equal to the gain in the log posterior probability of h. Let r denote the hypothesis that the signal was unintended, so that $P(r) = \rho$.

We state the following facts:

$$
\begin{aligned}
P(D|\neg h, r) = P(D|\neg h, \neg r) &= P(D|\neg h) \\
P(D|\neg h) &\approx P(D), \quad \text{when } P(h) \approx 0 \\
P(D|h, r) &= P(D), \quad \text{assuming that } D,\ h \text{ are independent conditioned on } r
\end{aligned}
$$

The likelihood under the true hypothesis is

$$P(D|h) = \rho P(D|h, r) + (1-\rho) P(D|h, \neg r)$$

We consider the case where $P >> \sigma$ so that the components of the mixture are well-separated. This means that the contribution of the false component is negligible and that we can approximate the likelihood $P(D|h)$ piecewise as follows:

$$P(D|h) \approx \begin{cases} P(D|h, r)P(r) & \text{if } D \sim P(D|h, r) \\ P(D|h, \neg r)P(\neg r) & \text{if } D \sim P(D|h, \neg r). \end{cases}$$

That is, when D is likely drawn from the unintended distribution, $P(D|h) \approx P(D|h, r)P(r)$. When D is likely drawn from the intended distribution, $P(D|h) \approx P(D|h, \neg r)P(\neg r)$.

$$
\begin{aligned}
E_{D|r,h}\left[\ln P(D|h)\right] &\approx E_{D|r,h}\left[\ln\Big(P(D|h, r)P(r)\Big)\right] \\
&= \ln \rho - h(D|h, r)
\end{aligned}
$$

and similarly

$$E_{D|\neg r,h}\left[\ln P(D|h)\right] \approx \ln(1-\rho) - h(D|h, \neg r)$$

The expected log likelihood of the data under the true hypothesis is

$$
\begin{aligned}
E_D\left[\ln P(D|h)\right] &= \rho E_{D|r,h}\left[\ln P(D|h)\right] + (1-\rho) E_{D|\neg r,h}\left[\ln P(D|h)\right] \\
&\approx \rho(\ln \rho - h(D|h, r)) + (1-\rho)(\ln(1-\rho) - h(D|h, \neg r)) \\
&= -H_2(\rho) - \rho h(D|h, r) - (1-\rho) h(D|h, \neg r) \\
&= -H_2(\rho) - \rho h(D) - (1-\rho) h(D|h, \neg r), \quad \text{assuming } D|r,\ h|r, \text{ independent}
\end{aligned}
$$

Where H_2 is the binary entropy function.

The likelihood of the alternative hypothesis is

$$E_{D|h}\left[\ln P(D|\neg h)\right] \approx E_{D|h}\left[\ln P(D)\right]$$

To simplify this term we use its expectation over h. Recalling the fact that,

$$E_a\left[E_{b|a}[f(b)]\right] = E_b[f(b)]$$

, we find that this term's expectation over h is $-h(D)$.

So the total expected gain in the logits of our hypothesis is

$$E_{D|h}[\ln P(D|h) - P(D|\neg h)] \approx -H_2(\rho) + (1-\rho)h(D) - (1-\rho)h(D|h, \neg r)$$

In our particular model, D is uniformly distributed over an interval of length P and the entropy of D conditional on h is just the entropy of the gaussian noise. Denoting the latter term by H_σ, we have,

$$E_{D|h}[\ln P(D|h) - P(D|\neg h)] \approx -H_2(\rho) + (1-\rho)(\ln P - H_\sigma)$$

This has a nice interpretation: we gain $\ln P - H_\sigma$ bits of information per intended signal, but the presence of unintended signals makes us pay a penalty in two ways: only $(1-\rho)$ of the time is our signal intended, and additionally we must expend $H_2(\rho)$ bits to learn whether the signal was intended.

Unfortunately the singularity of the binary entropy function at $\rho = 0$ becomes a practical problem for fast text entry. For example, a seemingly low outlier rate of $\rho = 0.03$ imposes the relatively high cost of $H_2(0.03) \approx 0.19$ bits in addition to the 3% loss that would be naively expected. Under reasonable parameters for an intermediate user (e.g. $\sigma = 25ms$, $P = 700ms$), these combined terms amount to a loss of more than 10% of the channel's capacity. Thus to achieve expert entry speeds, it is necessary to make unintended signals very rarely.

4 Embellishments

4.1 Branches of the Tree

References

[1] Artemy Kolchinsky and Brendan Tracey. Estimating mixture entropy with pairwise distances. *Entropy*, 19(7):361, July 2017.